

Theoretical Physics 2 - Blatt 05

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13. Bra-Ket Notation Consider the states $|\psi_n\rangle$ as eigenstates of a Hermitian operator H with

$$H|\psi_n\rangle = \lambda_n |\psi_n\rangle$$

Assume that the $|\psi_n\rangle$ form a discrete orthonormal basis.

1. The operator U_{mn} is defined as

$$U_{mn} = |\psi_m\rangle \langle \psi_n|.$$

Compute the adjoint operator $U_{mn}^\dagger$, the commutator $[U_{mn}, H]$, and show that

$$U_{mn}U_{pq} = \delta U_{mq}$$

(kronecker delta).

Calculation

Calculation

$$U_{mn}^\dagger = (|\psi_m\rangle \langle \psi_n|)^\dagger = \langle \psi_n|^\dagger |\psi_m\rangle^\dagger = |\psi_n\rangle \langle \psi_m| = U_{nm}$$

Calculation

$$U_{mn}U_{pq} = |\psi_m\rangle \langle \psi_n|\psi_p\rangle \langle \psi_q| = |\psi_m\rangle \delta_{np} \langle \psi_q| = \delta_{np} |\psi_m\rangle \langle \psi_q| = \delta_{np} U_{mq}$$

2. Show that for an arbitrary operator A and

$$a_{mn} = \langle \psi_m|A|\psi_n\rangle,$$

the following holds:

$$A = \sum_{m,n} a_{mn} U_{mn}$$

Calculation

Lets define $B := \sum_{m,n} a_{mn} U_{mn}$ and let B operate on an arbitrary basis vector $|\psi_k\rangle$:

$$B |\psi_k\rangle = \sum_{m,n} a_{mn} |\psi_m\rangle \langle \psi_n | \psi_k \rangle = \sum_{m,n} a_{mn} |\psi_m\rangle \delta_{nk} = \sum_m a_{mk} |\psi_m\rangle = \sum_n \langle \psi_m | A | \psi_k \rangle |\psi_m\rangle = \sum_n |\psi_m\rangle \langle \psi_m | A | \psi_k \rangle = A |\psi_k\rangle$$

Calculation

$$|\phi\rangle = \sum_m |\psi_m\rangle \langle \psi_m | \phi \rangle$$

Therefore $a = B = \sum_{m,n} a_{mn} U_{mn}$

or other way of calculation

Calculation

Completeness relation $I = \sum_m |\psi_m\rangle \langle \psi_m|$

$$A = I A I = \left(\sum_m |\psi_m\rangle \langle \psi_m| \right) A \left(\sum_n |\psi_n\rangle \langle \psi_n| \right) = \sum_{m,n} |\psi_m\rangle \langle \psi_m | A | \psi_n \rangle \langle \psi_n| = \sum_{m,n} |\psi_m\rangle a_{mn} \langle \psi_n| = \sum_{m,n} a_{mn} |\psi_m\rangle \langle \psi_n| = \sum_{mn} a_{mn} U_{mn}$$

3. Consider the following states:

$$|\phi\rangle = \frac{1}{\sqrt{2}} |\psi_1\rangle + \frac{i}{2} |\psi_2\rangle + \frac{1}{2} |\psi_3\rangle,$$

$$|X\rangle = \frac{1}{4} |\psi_1\rangle + \frac{1}{3} |\psi_3\rangle.$$

Are these states normalized?

Calculation

$$\langle \phi | \phi \rangle \stackrel{!}{=} 1$$

Calculation

$$\langle \phi | \phi \rangle = \left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{i}{2} \right|^2 + \left| \frac{1}{2} \right|^2 = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$$

$$\langle X | X \rangle \stackrel{!}{=} 1$$

Calculation

$$\langle X | X \rangle = \left| \frac{1}{4} \right|^2 + \left| \frac{1}{3} \right|^2 = \frac{1}{16} + \frac{1}{9} \neq 1$$

Are they orthogonal?

Calculation

$$\langle \phi | X \rangle \stackrel{!}{=} 0$$

Calculation

$$\begin{aligned} \langle \phi | X \rangle &= |\phi\rangle^\dagger |X\rangle = \left(\frac{1}{\sqrt{2}} |\psi_1\rangle + \frac{i}{2} |\psi_2\rangle + \frac{1}{2} |\psi_3\rangle\right)^\dagger \left(\frac{1}{4} |\psi_1\rangle + \frac{1}{3} |\psi_3\rangle\right) \\ &= \left(\frac{1}{\sqrt{2}} \langle \psi_1| - \frac{i}{2} \langle \psi_2| + \frac{1}{2} \langle \psi_3|\right) \left(\frac{1}{4} |\psi_1\rangle + \frac{1}{3} |\psi_3\rangle\right) = \frac{1}{\sqrt{2}} \frac{1}{4} + \frac{1}{2} \frac{1}{3} > 0 \end{aligned}$$

Compute the projectors onto $|\phi\rangle$ and $|X\rangle$, and show that the projectors are self-adjoint operators.

Calculation

$$\text{Let } |\psi_1\rangle := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |\psi_2\rangle := \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |\psi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Calculation

$$\begin{aligned} |\phi\rangle &= \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{2} \\ \frac{1}{2} \end{pmatrix}, \langle \phi| = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{2} & \frac{1}{2} \end{pmatrix} \\ P_\phi &= |\phi\rangle \langle \phi| = \begin{pmatrix} \frac{1}{2} & -\frac{i}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \\ \frac{i}{2\sqrt{2}} & \frac{1}{4} & \frac{i}{4} \\ \frac{1}{2\sqrt{2}} & -\frac{i}{4} & \frac{1}{4} \end{pmatrix} \end{aligned}$$

Calculation

$$\begin{aligned} |\phi\rangle &= \begin{pmatrix} \frac{1}{4} \\ 0 \\ \frac{1}{3} \end{pmatrix}, \langle \phi| = \begin{pmatrix} \frac{1}{4} & 0 & \frac{1}{3} \end{pmatrix} \\ P_\phi &= |\phi\rangle \langle \phi| = \begin{pmatrix} \frac{1}{16} & 0 & \frac{1}{12} \\ 0 & 0 & 0 \\ \frac{1}{12} & 0 & \frac{1}{9} \end{pmatrix} \end{aligned}$$

4. Let $|\phi\rangle$ and $|X\rangle$ be two arbitrary vectors in the state space. Under what condition is

$$K = |\phi\rangle \langle X|$$

Hermitian?

Calculation

Condition $K^\dagger \stackrel{!}{=} K$

Calculation

$$K^\dagger = |X\rangle \langle \phi|$$

That does not hold for all $|\phi\rangle, |X\rangle$. That can only be true if they are linear dependent
 $|X\rangle = \lambda |\phi\rangle$

Calculation

$$K = |\phi\rangle \langle X| = |\phi\rangle (\lambda |\phi\rangle)^\dagger = |\phi\rangle \lambda^* \bar{\phi} = \lambda^* |\phi\rangle \bar{\phi}$$

$$\implies \lambda = \lambda^*$$

Under what condition is K a projector?

Calculation

$$K^2 \stackrel{!}{=}$$

Calculation

$$K^2 = (|\phi\rangle \langle X|)(|\phi\rangle \langle X|) = |\phi\rangle \langle X|\phi\rangle \langle X| = \langle X|\phi\rangle |\phi\rangle \langle X| = \langle X|\phi\rangle K$$

$$\implies$$

Calculation

Trivial for $K = 0$

$$\langle X|\phi\rangle K = K \implies \langle X|\phi\rangle = 1$$

Show that K can always be written as

$$K = \alpha P_1 P_2$$

where P_i are projectors and $\alpha \in \mathbb{C}$.

Calculation

Start with projections in the direction of $|\phi\rangle$ and $|X\rangle$

Calculation

For $ket\phi \neq 0$ and $|X\rangle \neq 0$

we get the normalized

$$|\hat{\phi}\rangle = \frac{|\phi\rangle}{\|\phi\|} \text{ and } |\hat{X}\rangle = \frac{|X\rangle}{\|X\|}$$

Let $P_1 = |\hat{\phi}\rangle \langle \hat{\phi}|$, $P_2 = |\hat{X}\rangle \langle \hat{X}|$ and calculate:

Calculation

$$P_1 P_2 = |\hat{\phi}\rangle \langle \hat{\phi}| \hat{X} \rangle \langle \hat{X}| = \langle \hat{\psi} | \hat{X} \rangle |\hat{\phi}\rangle \langle \hat{X}|.$$

$\Rightarrow$

$$|\hat{\phi}\rangle \langle \hat{X}| = \frac{1}{\|\phi\| \|\hat{X}\|} |\phi\rangle \langle X|$$

$$P_1 P_2 = \frac{\langle \hat{\phi} | \hat{X} \rangle}{\|\phi\| \|\hat{X}\|} |\phi\rangle \langle X|$$

With this $K = |\phi\rangle \langle X| = \frac{\|\phi\| \|\hat{X}\|}{\langle \hat{\phi} | \hat{X} \rangle} P_1 P_2 = \frac{\|\phi\|^2 \|\hat{X}\|^2}{\langle \hat{\phi} | \hat{X} \rangle} P_1 P_2 = \alpha P_1 P_2$

5. Let $H |\phi_n\rangle = \lambda_n |\phi_n\rangle$ and assume that the eigenvalues λ are non-degenerate. Show that the so-called **null operator**:

$$\Pi_0 = \prod_n (H - \lambda_n)$$

satisfies

$$\Pi_0 |\phi\rangle = 0$$

for all vectors $|\phi\rangle$.

Calculation

Want to show, that when $|\phi_n\rangle$ is as basis, that $\Pi_0 = 0$ on any basis vectors.

Calculation

Take an arbitrary eigenvector $|\phi_m\rangle$ and let's apply $H - \lambda_n$ on it:

$$(H - \lambda_n) |\phi_m\rangle = (\lambda_m - \lambda_n) |\phi_m\rangle,$$

Let's look at the null operator

Calculation

$$\Pi_0 |\phi_m\rangle = \prod_n (H - \lambda_n) |\phi_m\rangle = \prod_n (\lambda_m - \lambda_n) |\phi_m\rangle = \dots \cdot (\lambda_m - \lambda_m) \cdot \dots = 0$$

and applied to any vector $|\phi\rangle = \sum_m c_m |\phi_m\rangle$

Calculation

$$\Pi_0 |\phi\rangle = \Pi_0 \sum_m c_m |\phi_m\rangle = \sum_m c_m \Pi_0 |\phi_m\rangle = \sum_m c_m \cdot 0 = 0$$

How does the operator

$$\prod_{n \neq m} \left(\frac{H - \lambda_n}{\lambda_m - \lambda_n} \right)$$

act?

Calculation

Let $P_m := \prod_{n \neq m} \left(\frac{H - \lambda_n}{\lambda_m - \lambda_n} \right)$
 Lets apply P_m an an arbitrary eigenvector $|\phi_k\rangle$

Calculation

$P_m |\phi\rangle$
 Case 1: $k = m$

Calculation

$P_m |\phi_m\rangle = \prod_{n \neq m} \left(\frac{\lambda_m - \lambda_n}{\lambda_m - \lambda_n} |\phi_m\rangle \right)$
 $\Rightarrow$
 $P_m |\phi_m\rangle = 1 |\phi_m\rangle$

Case 2: $k \neq m$

Calculation

$\lambda_k - \lambda_k = 0$ Therefore $P_m |\phi_k\rangle = 0$

For any arbitrary vector $|\phi\rangle = \sum_k c_k |\phi_k\rangle$ the operator takes only the m-th component of the vector.

$P_m |\phi\rangle = \sum_k c_k P_m |\phi_k\rangle \sum_k \delta_{mk} |\phi_k\rangle = c_m |\phi_m\rangle$

Can this operator be expressed in another way?

Calculation

$P_m = |\phi_m\rangle \langle \phi_m|$
 $|\phi_m\rangle \langle \phi_m| |\phi_k\rangle = \delta_{mk} |\phi_m\rangle$
 Same effect, because it takes the k-th component of the vector $|\phi\rangle$

6. Compute the matrix elements of the momentum operator $\hat{p}$ in the position representation,

$$\langle x' | \hat{p} | x \rangle,$$

using the momentum representation

$$\langle p' | p \rangle = \delta(p' - p), \tag{1}$$

and the known wavefunction

$$\langle x|p\rangle = \frac{\exp \frac{ipx}{\hbar}}{\sqrt{2\pi\hbar}}$$

Calculation

$$1 = \int dp |p\rangle \langle p|$$

$$\langle x'|1 \cdot \hat{p}|x\rangle = \int dp' \int dp \langle x'|p'\rangle \langle p'|\hat{p}|p\rangle \langle p|x\rangle$$

Calculation

Eigenstate/ eigenvector $|p\rangle$

$$\hat{p}|p\rangle = p|p\rangle \implies \langle p'|\hat{p}|p\rangle = p \langle p'|p\rangle = p\delta(p' - p)$$

Plug in $\langle x'|\hat{p}|x\rangle = \int dp' \int dp \langle x'|p'\rangle p\delta(p' - p) \langle p|x\rangle = \int dp \langle x'|p\rangle p \langle p|x\rangle$
 plug in wavefunction and adjungated wavefunction ($\langle p|x\rangle = \frac{\exp \frac{-ipx}{\hbar}}{\sqrt{2\pi\hbar}}$)

$$\langle x'|\hat{p}|x\rangle = \int dp \frac{\exp \frac{ipx'}{\hbar}}{\sqrt{2\pi\hbar}} p \frac{\exp \frac{-ipx}{\hbar}}{\sqrt{2\pi\hbar}} = \frac{1}{2\pi\hbar} \int dp p \exp \frac{ip(x'-x)}{\hbar}$$

Using Identity: $\delta(x' - x) = \frac{1}{2\pi\hbar} \int dp \exp \frac{ip(x'-x)}{\hbar}$

the derivation of the Identity $\frac{\partial}{\partial x'} \delta(x' - x) = \frac{1}{2\pi\hbar} \int dp \frac{ip}{\hbar} \exp \frac{ip(x'-x)}{\hbar}$

And therefore $\frac{1}{2\pi\hbar} \int dp p \exp \frac{ip(x'-x)}{\hbar} = -i\hbar \frac{\partial}{\partial x'} \delta(x' - x)$

Result $\langle x'|\hat{p}|x\rangle = -i\hbar \frac{\partial}{\partial x'} \delta(x' - x)$

Two-dimensional Harmonic Oscillator: Consider the two-dimensional motion of a particle of mass m in a potential of the form

$$V(r) = \frac{1}{2}m\omega^2 r^2, \text{ with } r = \sqrt{x^2 + y^2}$$

1. Consider the Hamiltonian operator $\hat{H}$ of the system and show that it can be written as a sum

$$\hat{H} = \hat{H}_x + \hat{H}_y,$$

where $\hat{H}_x$ and $\hat{H}_y$ are Hamiltonians of one-dimensional harmonic oscillators.

Calculation

$$\begin{aligned}
 T &= \frac{1}{2m}(p_x^2 + p_y^2) \\
 V &= \frac{1}{2}m\omega^2(x^2 + y^2) \\
 H &= T + V \\
 H_x &= \frac{1}{2m}p_x^2 + \frac{1}{2}m\omega^2x^2 \\
 H_y &= \frac{1}{2m}p_y^2 + \frac{1}{2}m\omega^2y^2 \\
 H &= H_x + H_y
 \end{aligned}$$

Introduce the raising and lowering operators $\hat{a}_x, \hat{a}_x^\dagger$ and $\hat{a}_y, \hat{a}_y^\dagger$ for these oscillators and express $\hat{H}$ in terms of them.

Calculation

$$\begin{aligned}
 a_x &= \frac{1}{\sqrt{2\hbar m\omega}}(m\omega x + ip_x) \\
 a_x^\dagger &= \frac{1}{\sqrt{2\hbar m\omega}}(m\omega x - ip_x) \\
 a_y &= \frac{1}{\sqrt{2\hbar m\omega}}(m\omega y + ip_y) \\
 a_y^\dagger &= \frac{1}{\sqrt{2\hbar m\omega}}(m\omega y - ip_y) \\
 H_x &= \hbar\omega(a_x^\dagger a_x + \frac{1}{2}) \\
 H_y &= \hbar\omega(a_y^\dagger a_y + \frac{1}{2}) \\
 H &= \hbar\omega(a_y^\dagger a_y + \frac{1}{2})
 \end{aligned}$$

What follows for the energy eigenvalues of $\hat{H}$? Express the corresponding energy eigenfunctions using the ground state

$$|\phi_{00}\rangle = |\phi_0^x, \phi_0^y\rangle$$

(where $\hat{a}_x |\phi_0^x\rangle = 0$ and $\hat{a}_y |\phi_0^y\rangle = 0$) and the raising operators $\hat{a}_x^\dagger$ and $\hat{a}_y^\dagger$.

Calculation

$$n_x, n_y \in \mathbb{N}_0$$

$\implies$

$$E = \hbar\omega(n_x + n_y + 1)$$

Calculation

$$|\phi_{00}\rangle = |\phi_0^x, \phi_0^y\rangle$$

$$\text{with } a_x |\phi_0^x\rangle = 0, a_y |\phi_0^y\rangle = 0$$